

A New Hybrid Algorithm Based on black hole Optimization and Bisecting K-means for Cluster Analysis

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Abstract — Clustering is a popular data analysis and data mining technique. In bisecting k-means clustering technique, the data is incrementally partitioned into K clusters. However, the performance of bisecting k-means algorithm highly depends on the initial state and it may converge to a local optimum solution. To solve these problems, a hybrid evolutionary algorithm using combination of BH (black hole) and bisecting k-means algorithms, called BH-BKmeans is proposed. With this, a dataset would be precisely clustered in a reasonable time complexity and led to global optimum with local refine in clustering. The performance of the proposed algorithm is evaluated through several benchmark data sets. The simulation results show that the proposed algorithm outperforms other typical clustering algorithms such as bisecting k-means, BH and PSO.

Index Terms – Bisecting k-means, Black hole, data clustering, Nature-inspired algorithm.

I. Introduction

Data clustering grouping data into clusters such that the data in each cluster have a high degree of similarity while being very dissimilar to data from other clusters. Dissimilarities are assessed according to the attribute values describing the objects. Data clustering has roots in a number of areas; including data mining, machine learning, biology, and statistics[1]. Bisecting k-Means clustering algorithm is one of the most popular partitional clustering used in variety of domains. The Bisecting k-means algorithm is defined over continuous data[2]. The Bisecting k-means algorithm gave better results only when the initial partitions were close to the final solution. In other words, the results of Bisecting k-means highly depend on the initial state and reach to local optimal solution[1].

On the other hand, black hole algorithm is a new natural inspired algorithm for dealing with complex and optimization problems which performs better than other traditional heuristic algorithms. Also, it overcomes local optimal solution [2]. In

contrast, black hole algorithm unable to refine centre of clusters as fast as other algorithms like k-Means.

So, the combination of K-means method and black hole in a bisecting manner has high potential to find the cluster centers in contrast to single methods. In this paper, a new clustering algorithm using a combination of conventional bisecting K-means algorithm and the black hole called BH-BKmeans. In section 2, 3 and 4 the basic methods are briefly reviewed. Section 5 addresses the proposed hybrid method. In section 6, the proposed method is validated by comparing to other well-known methods in a right experimental framework. Last section includes the conclusion.

II. k-means algorithm

k-means is a popular algorithm that partitions a data set into k clusters[1]. the data set object showed by $d_1; d_2; \dots; d_n$, then the clustering is the process of grouping object into k clusters so that the global criterion function is either minimized or maximized,

$$\sum_{j=1}^k \sum_{i=1}^n \text{sim}(d_i, c_j) \quad (1)$$

depending on the definition of $\text{sim}(d_i; c_j)$. C_j represents the centroid of cluster C_j , for $j: 1; \dots; k$, and $\text{sim}(d_i; c_j)$ evaluates the similarity between a document d_i and a centroid c_j . When the vector space model is used to represent the object and the cosine is used for $\text{sim}(d_i; c_j)$ each object is assigned to the cluster whose centroid vector is more similar to the object than those of other clusters, and the global criterion function is maximized in that case. This optimization process is known as an NP-complete problem [3], and the k-means algorithm was proposed to provide an approximate solution. The steps of k-means are as follows:

1. Select k initial cluster centroids.
2. compute the similarity with each cluster centroid, and assign the document to the closest (i.e., most similar) centroid. (assignment step)

3. Recalculate k centroids based on the documents assigned to them.
4. Repeat steps 2 and 3 until convergence[4].

III. Bisecting k-means algorithm

The bisecting k-means algorithm [5] is a variant of k-means. In the bisecting k-means only one cluster is split into two subclusters at each step. This algorithm starts with the whole data set as a single cluster, and its steps are as follows[6]:

1. Select a cluster C_j to split based on a heuristic function.
2. Find 2 subclusters of C_j using the k-means algorithm: (bisecting step)
 - (a) Select 2 initial cluster centroids.
 - (b) For each object of C_j , compute the similarity with the 2 cluster centroids, and assign the object to the closer centroid. (assignment step)
 - (c) Recalculate 2 centroids based on the objects assigned to them.
 - (d) Repeat steps 2b and 2c until convergence.
3. Repeat step 2 I times, and select the split that produces the best clustering result in terms of the global criterion function.
4. Repeat steps 1, 2 and 3 until k clusters are obtained.

IV. Black hole algorithm

The BH algorithm is a population-based method and has some common features with other population-based methods[2]. As with other population-based algorithms, a population of candidate solutions to a given problem is generated and distributed randomly in the search space. The population-based algorithms evolve the created population towards the optimal solution via certain mechanisms. For example, in GAs, the evolving is done by mutation and crossover operations[7]. In PSO, this is done by moving the candidate solutions around in the search space using the best found locations[8], which are updated as better locations are found by the candidates. The ant colony optimization (ACO) was inspired from the behavior of a real ant colony, which is able to find the shortest path between its nest and a food source [9]. The intelligent water drops (IWDs) algorithm was inspired from observing natural water drops that flow in rivers and how natural rivers find almost optimal paths to their destination[10]. The firefly algorithm (FA) was inspired by the flashing behavior of fireflies in nature [11], while the honey bee mating optimization (HBMO) algorithm was inspired by the process of marriage in real honey bees [12]. The harmony search was inspired by the improvising process of composing a piece of music. The action of finding the harmony in music is similar to finding the optimal solution in an optimization process [13]. In the BH algorithm the evolving of the population is done by moving all the candidates towards the best candidate in each iteration, namely, the black hole and replacing those candidates that enter within the range of the black hole by newly generated candidates in the search space. The BH algorithm in this paper is more similar to the natural

black hole phenomenon and is completely different from the black hole PSO in terminology which has been used for the first time in solving benchmark functions [2]. In BH algorithm the best candidate among all the candidates at each iteration is selected as a black hole and all the other candidates form the normal stars. The creation of the black hole is not random and it is one of the real candidates of the population. Then, all the candidates are moved towards the black hole based on their current location and a random number. The details of the BH algorithms are as follows[2]:

Like other population-based algorithms, in the black hole algorithm a randomly generated population of candidate solutions – the stars – are placed in the search space of some problem or function. After initialization, the fitness values of the population are evaluated and the best candidate in the population, which has the best fitness value, is selected to be the black hole and the rest form the normal stars. The black hole has the ability to absorb the stars that surround it.

After initializing the black hole and stars, the black hole starts absorbing the stars around it and all the stars start moving towards the black hole. The absorption of stars by the black hole is formulated as follows[2]:

$$x_i(t+1) = x_i(t) + rand \times (x_{BH} - x_i(t)) \quad i = 1, 2, \dots, N \quad (2)$$

where $x_i(t)$ and $x_i(t+1)$ are the locations of the i th star at iterations t and $t+1$, respectively. x_{BH} is the location of the black hole in the search space. $rand$ is a random number in the interval $[0, 1]$. N is the number of stars (candidate solutions).

While moving towards the black hole, a star may reach a location with lower cost than the black hole. In such a case, the black hole moves to the location of that star and vice versa. Then the BH algorithm will continue with the black hole in the new location and then stars start moving towards this new location.

In addition, there is the probability of crossing the event horizon during moving stars towards the black hole. Every star (candidate solution) that crosses the event horizon of the black hole will be sucked by the black hole. Every time a candidate (star) dies – it is sucked in by the black hole – another candidate solution (star) is born and distributed randomly in the search space and starts a new search. This is done to keep the number of candidate solutions constant. The next iteration takes place after all the stars have been moved.

The radius of the event horizon in the black hole algorithm is calculated using the following equation[2]:

$$R = \frac{f_{BH}}{\sum_{i=1}^N f_i} \quad (3)$$

where f_{BH} is the fitness value of the black hole and f_i is the fitness value of the i th star. N is the number of stars (candidate solutions).

When the distance between a candidate solution and the black hole (best candidate) is less than R , that candidate is collapsed

and a new candidate is created and distributed randomly in the search space.

Based on the above description the main steps in the BH algorithm are summarized as follows[2]:

```

Initialize a population of stars with random locations in the search
space
Loop
  For each star, evaluate the objective function
  Select the best star that has the best fitness value as the black
  hole
  Change the location of each star according to Eq. (2)
  If a star reaches a location with lower cost than the black hole,
  exchange their locations
  If a star crosses the event horizon of the black hole, replace it with
  a new star in a random location in the search space
  If a termination criterion (a maximum number of iterations or a
  sufficiently good fitness) is met, exit the loop
End loop

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According to [14] while the quantity of clusters goes beyond three the clustering problem becomes NP-hard.

The candidate solution to the clustering problem corresponds to a 1-dimensional array while applying black hole algorithm for data clustering. Every candidate solution is considered as k initial cluster centers and the individual unit in the array as the cluster center dimension. Fig.1 illustrates a candidate solution of a problem with three clusters and all the data objects have four features[2].

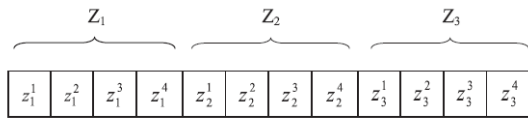


Fig. 1. Example of a candidate solution.

V. BH-BK-Means proposed algorithm

As we know, the BK-means algorithm result is sensitive to the selection of the initial cluster centroids and the outlier data points. A small number of such outlier data may cause BK-means algorithm converge to the local optima and substantially influence the final cluster results [1]. The initial selection of the cluster centroids decides the main processing of BK-means and the partition result of the dataset as well. The main processing of BK-means is to search the local optimal solution in the vicinity of the initial solution and to refine the partition results. If the right initial clustering centroids can be obtained using any of the other techniques, the BK-means would work well in refining the clustering centroids to find the optimal clustering centers. It is necessary to employ some other global optimal searching algorithm for generating the initial cluster centroids [17]. The black hole algorithm can be used to generate good initial cluster centroids for the BK-means. In this paper, we present a hybrid BH+BK-

means clustering algorithm that performs a faster clustering and can avoid being trapped in a local optimal solution.

The hybrid BH+BK-means algorithm includes two modules, the BH module and the BKmeans module. The global searching stage and local refine stage are accomplished by these two modules, respectively. In the initial stage, the BH module is executed for a short period to discover the vicinity of the optimal solution by a global search and at the same time to avoid consuming high computation. The result from the BH module is used as the initial seed for the BK-means module. The BK-means algorithm is applied to refine and generate the final results. The whole approach can be summarized as:

Loop

- (1) Starting the BH clustering process until the maximum number of iterations is exceeded with $k = 2$
- (2) Inheriting clustering result from BH as the initial centroid vectors of BK-means module.
- (3) Starting BK-means process until maximum number of iterations is reached.
- (4) find worse cluster and repeat algorithm for it.
- (5) When a maximum number of cluster obtain exit the loop.

End loop

In the BH+BK-means algorithm, the ability of the global searching of the BH algorithm and the fast convergence of the K-means algorithm are combined. The BH algorithm is used at the initial stage to help discovering the vicinity of the optimal solution by a global search. The result from BH is used as the initial seed of the BK-means algorithm, which is applied for refining and generating the final result.

Some researchers (Omran, Engelbrecht, & Salman, 2005; Merwe & Engelbrecht, 2003) argue that the BK-means algorithm tends to converge faster than other clustering algorithms, but usually with less accuracy. The performance of the clustering algorithm can be improved by seeding the initial swarm with the result of the K-means algorithm.

The experimental results illustrates that the sequence of the hybrid BH-BK-means algorithms is very important. Performing the BK-means clustering in advance of the BH clustering module will reduce the global searching ability of the BH algorithm and reduce the whole algorithm's performance.

VI. Experimental result

Four benchmark datasets with a variety of complexity are used to evaluate the performance of the proposed approach. The datasets are Iris, Glass Vowel and Contraceptive Method Choice (CMC), which are available in the repository of the machine learning databases [15]. Table 1 summaries the main characteristics of the used datasets.

The performance of the BH-BK-Means algorithm is compared against well known and the most recent algorithms reported in the literature, including BK-means [16], particle swarm optimization [8] and the black hole algorithm [2].

The performance of the algorithms is evaluated and compared using two criteria:

- Sum of intra-cluster distances as an internal quality measure: The distance between each data object and the center of the corresponding cluster is computed and summed up, as defined in Eq. (4).

$$F(O, Z) = \sum_{i=1}^N \sum_{j=1}^K w_{ij} \|O_i - Z_j\|^2 \quad (4)$$

Table 1

Main characteristics of the test datasets.

Dataset	Number of clusters	Number of features	Number of data objects
Iris	3	4	150 (50,50,50)
Glass B	6	9	214 (70, 76,17, 13, 9,29)
Vowel	6	3	871 (72, 89,172,151,207,180)
CMC	3	9	1473 (629,334,510)

Table 2

The sum of intra-cluster distances obtained by algorithms on different datasets.

Dataset	Criteria	BK-means	PSO	BH	BH-BK-means
Iris	Best	97.34572	96.87935	96.65589	79.154
	Average	103.72902	98.14236	96.65681	84.137
	Worst	126.40420	99.76952	96.66306	92.114
Glass	Best	214.67753	223.90546	210.51549	171.453
	Average	229.97785	230.49328	211.49860	189.219
	Worst	261.87819	246.08915	213.95689	210.515
Vowel	Best	147,394.80398	152,461.56473	148,985.61373	122,145.11
	Average	151,660.80712	153,218.23418	149,848.18144	138,985.613
	Worst	167,474.26593	158,987.08231	153,058.98663	148,915.71
CMC	Best	5539.18214	5539.17452	5532.88323	2832.791
	Average	5541.42344	5547.89320	5533.63122	3342.883
	Worst	5545.33338	5561.65492	5534.77738	5352.883

where $\|O_i - Z_j\|$ is the Euclidean distance between a data object O_i and the cluster center Z_j . N and K are the number of data objects and the number of clusters, respectively. w_{ij} is the association weight of data object O_i with cluster j , which will be either 1 or 0 (if object i is assigned to cluster j ; w_{ij} is 1, otherwise 0).

Fuzzy clustering allows w_{ij} to take values in the interval (0, 1). Clearly, the smaller the sum of intra-cluster distances, the higher the quality of the clustering. The sum of intra-cluster distances is also the evaluation fitness in this work.

- Error Rate (ER) as an external quality measure: The percentage of misplaced data objects, as shown in the following equation:

$$ER = \frac{\text{number of misplaced objects}}{\text{total number of objects within dataset}} \times 100$$

A summary of the intra-cluster distances obtained by the clustering algorithms is given in Table 2. The values reported are best, average, worst and the standard deviation of solutions over 50 independent simulations.

As seen from the results in Table 2, the hybrid BH-BK-means algorithm achieved the best results among all the algorithms. For the Iris dataset, the best, worst, and average solutions

Table 3

The error rate of clustering algorithms on the test datasets

Dataset	BK-means(%)	PSO (%)	BH(%)	BH-BK-means(%)
Iris	12.42	10.06	10.02	09.95
Glass	37.46	41.20	36.51	32.14
Vowel	44.59	42.39	41.65	40.27
CMC	54.41	54.50	54.39	53.46

obtained by BH-BK-means are 79.154, 92.114 and 84.137, respectively, which are better than the other algorithms.

Table 3 shows the mean error rate obtained by the clustering algorithms from 50 simulation runs on the test datasets. As seen from the results in Table 3, the BH algorithm provided a minimum average error rate in all the test datasets.

VII. Conclusion

In this paper, a hybrid method for clustering with high accuracy is presented. For this purpose, the k-means and black hole algorithms are combined in a bisecting manner.

The combination of these algorithms has led to find global optimum with local refine in clustering. The result of experiments on various on real datasets shows that the new hybrid bisecting k-means and black hole algorithm can cause higher performance in finding center of clusters in compare to single ones.

VIII. Future work

Possible solutions to the problems of the classical clustering methods using classical methods and the integration of swarm intelligence.

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